NATIONAL UNIVERSITY OF COMPUTER AND EMERGING SCIENCES

Marks: 20	**QUIZ-4**	**Time: 20 Minutes**

Subject: Probability and Statistics **Section: G**
Instructor: Hazber Samson Date: 02-04-2026
Student Name_____ Roll No_____

Q1. **(3+3+2+2=10)**

An archer hits a bull's-eye with a probability of 0.09, and the results of different attempts can be taken to be independent of each other
(a) If the archer shoots a series of arrows, what is the probability that the *first* bull's-eye is scored with the fourth arrow?
(b) What is the probability that the *third* bull's-eye is scored with the tenth arrow?
(c) What is the expected number of arrows shot before the *first* bull's-eye is scored?
(d) What is the expected number of arrows shot before the *third* bull's-eye is scored?

Q2. **(3+3+4=10)**

The ages of prime ministers of Sri Lanka, when they were first sworn to office, are normally distributed with a mean of 58.21 years and a standard deviation of 11.56 years. Find the probability that the age of a randomly selected prime minister of Sri Lanka when he was sworn to office was
a. less than 50 years,
b. more than 70 years.
c. between 55 and 60 years.

GOOD LUCK !